

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using content-based filtering approach	
Experiment No: 11	Date:	Enrolment No: 92410133034

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.

In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:

In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.

Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:

The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

Advantages and Disadvantages:

- **Advantages:**
 - No need for data on other users when applying to similar users.
 - Able to recommend to users with unique tastes.
 - Able to recommend new & popular items

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- Explanations for recommended items.
- **Disadvantages:**
 - Finding the appropriate feature is hard.
 - Doesn't recommend items outside the user profile.

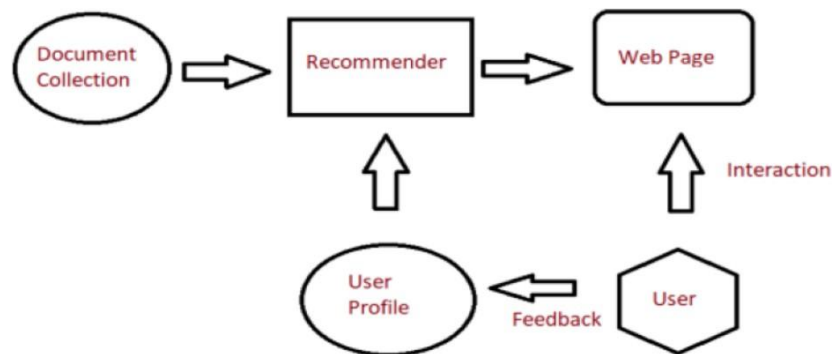


Fig: Recommender System

Pre Lab Exercise:

- When can you use content-based recommendation system approach?
 - When **item features are available** (like genre, keywords, description)
 - When recommendations depend on **user's past liked items**
 - When **other users' data is not available**
 - When users have **unique preferences**
 - When we want to recommend **similar items**
- Write advantages of content-based filtering approach
 - No need of data from other users
 - Works well for users with **unique tastes**
 - Can recommend **new and popular items**
 - Easy to **explain recommendations**
 - Personalized recommendations

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3. Write disadvantages of content-based filtering approach

- Difficult to **find correct features**
- Limited to **user profile only**
- Cannot recommend **new types of items (no diversity)**
- Requires **good feature extraction**
- Over-specialization problem

Program (Code):

```
[1]
✓ 1s
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Sample dataset (movies)
data = {
    'title': ['Movie A', 'Movie B', 'Movie C', 'Movie D'],
    'description': [
        'action adventure hero',
        'romantic love drama',
        'action thriller fight',
        'love romance emotional'
    ]
}

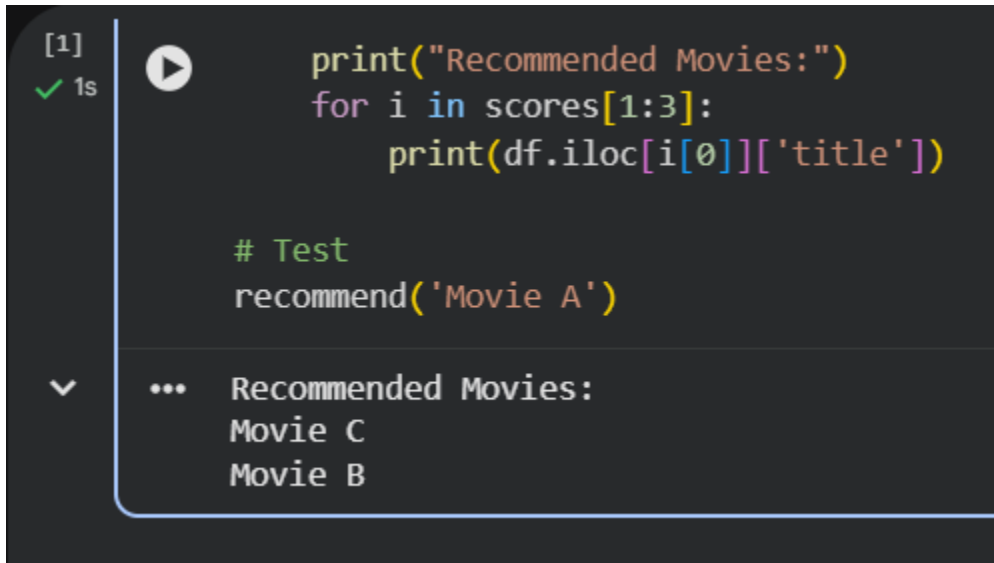
df = pd.DataFrame(data)

# Convert text to TF-IDF vectors
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(df['description'])

# Compute similarity
similarity = cosine_similarity(tfidf_matrix)

# Function to recommend movies
def recommend(movie_name):
    index = df[df['title'] == movie_name].index[0]
    scores = list(enumerate(similarity[index]))
    scores = sorted(scores, key=lambda x: x[1], reverse=True)
```

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```
[1] ✓ 1s
▶ print("Recommended Movies:")
  for i in scores[1:3]:
    print(df.iloc[i[0]]['title'])

# Test
recommend('Movie A')

... Recommended Movies:
Movie C
Movie B
```

Observation:

The content-based recommendation system was successfully implemented using TF-IDF vectorization and cosine similarity. The system analyzed the textual features (descriptions) of items and created item profiles. Based on these features, it calculated similarity between items and recommended those that are most similar to the selected item. The output shows that the system recommends relevant items according to user interest, demonstrating that content-based filtering works effectively when item features are properly defined. However, the system is limited to recommending items similar to those already liked and does not provide diverse suggestions beyond the user profile.

Post Lab Exercise:

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.

The system can be improved by using more advanced techniques and better data. Firstly, more features such as genre, cast, ratings, and reviews can be included to improve accuracy. Secondly, combining content-based filtering with collaborative filtering (hybrid system) can provide better recommendations. Also, using deep learning models can improve feature extraction from text. Additionally, improving data quality and increasing dataset size will enhance system performance. Finally, adding user feedback can help in making the system more personalized and adaptive.

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2. Recommend top-10 “series” using IMDB movie dataset

(Link: <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>). Attach the code with output. Also, write your observation for the same.

```
[8]
✓ 11s # Step 1: Upload file
from google.colab import files
uploaded = files.upload()

# Step 2: Import libraries
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Step 3: Load dataset (IMPORTANT: use exact file name shown after upload)
df = pd.read_csv('imdb_top_1000.csv')

# Step 4: Fill missing values (important for safety)
df['Genre'] = df['Genre'].fillna('')
df['Overview'] = df['Overview'].fillna('')

# Step 5: Combine features
df['content'] = df['Genre'] + " " + df['Overview']

# Step 6: Convert text to numbers
tfidf = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf.fit_transform(df['content'])

# Step 7: Similarity calculation
similarity = cosine_similarity(tfidf_matrix)

# Step 8: Recommendation function
def recommend(title):
    index = df[df['Series_Title'] == title].index[0]
    scores = list(enumerate(similarity[index]))
    scores = sorted(scores, key=lambda x: x[1], reverse=True)
```

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[8]

✓ 11s

▶

```

print("Top 10 Recommended Series:")
for i in scores[1:11]:
    print(df.iloc[i[0]]['Series_Title'])

# Step 9: Run example
recommend("The Dark Knight")

```

▼

⋮

Choose files

imdb_top_1000.csv

imdb_top_1000.csv(text/csv) - 438096 bytes, last modified: 16/04/2026 - 100% done

Saving imdb_top_1000.csv to imdb_top_1000.csv

Top 10 Recommended Series:

Batman Begins

The Dark Knight Rises

Kill Bill: Vol. 1

Joker

Batman: Mask of the Phantasm

La battaglia di Algeri

The Hurricane

Wreck-It Ralph

V for Vendetta

Kokuhaku